AI CAREER FOR WOMEN

Engineer Spoke

DL Based Image Caption Generator

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

by

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With Regards

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ABSTRACT

The Deep Learning-Based Image Caption Generator is an intelligent system developed to automatically generate meaningful and grammatically correct textual descriptions for images. With the rapid growth of digital images across digital platforms, manual image annotation has become inefficient and time-consuming. This project addresses this challenge by combining Computer Vision and Natural Language Processing techniques to automate caption generation.

The system is trained using the Flickr8K Dataset, which consists of 8,000 real-world images, each paired with five human-written captions. A hybrid deep learning architecture is implemented in which a Convolutional Neural Network (CNN) is used to extract rich visual features from images, and a Long Short-Term Memory (LSTM) network is used to generate captions sequentially. The CNN captures important objects and patterns within the image, while the LSTM learns language structure and predicts the next word in the sequence based on image features and previously generated words.

The model is evaluated using BLEU (Bilingual Evaluation Understudy) scores to measure the similarity between generated captions and reference captions. The results demonstrate satisfactory performance, indicating the system's ability to produce contextually relevant and human-like descriptions.

This project highlights the effective integration of deep learning models to bridge the gap between visual understanding and natural language generation. Furthermore, the system can be enhanced by incorporating attention mechanisms, transformer-based models, or larger datasets to improve caption accuracy and real-world applicability.

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CHAPTER 1

Introduction

1.1 Problem Statement:

With the rapid increase in digital images across social media, websites, and multimedia platforms, manually describing images has become time-consuming and inefficient. Traditional image processing techniques can detect objects but fail to understand the semantic meaning of an image or generate meaningful textual descriptions. There is a need for an intelligent automated system that can analyze an image, understand its content, and generate accurate and grammatically correct captions. Therefore, this project proposes a deep learning-based image caption generator that combines computer vision and natural language processing techniques to solve this problem.

1.2 Motivation:

The motivation behind this project is to bridge the gap between visual data and textual understanding. Image captioning has real-world applications in assistive technologies for visually impaired individuals, content management systems, search engines, and social media platforms. With advancements in deep learning, especially Convolutional Neural Networks (CNN) and Recurrent Neural Networks (LSTM), it has become possible to build systems that can understand images and generate human-like descriptions. This project explores the practical implementation of these technologies to create an efficient automated captioning system.

1.3 Objective:

The main objectives of this project are:

- To develop an automated system that generates meaningful captions for images.
- To use the Flickr8K Dataset for training and testing the model.
- To implement a hybrid deep learning architecture using CNN for feature extraction and LSTM for caption generation.
- To preprocess image and text data effectively for model training.
- To evaluate model performance using BLEU score metrics.
- To demonstrate the integration of Computer Vision and NLP techniques in a real-world application.

1.4 Scope of the Project:

The scope of this project is limited to generating captions for static images using a deep learning-based model trained on a benchmark dataset. The system focuses on generating short, descriptive sentences for general images. Future improvements may include using larger datasets such as MS COCO, implementing attention mechanisms, applying transformer-based models, and deploying the system as a web application. The project can also be extended to support real-time caption generation and assistive technologies for visually impaired users.

CHAPTER 2

Literature Survey

2.1 Review relevant literature

Image captioning has gained significant attention in the field of artificial intelligence due to its ability to bridge visual understanding and natural language generation. Early approaches primarily relied on template-based methods, where detected objects and attributes were inserted into predefined sentence structures. However, these methods were limited in their ability to generate natural and diverse descriptions. With advancements in deep learning, researchers began using Convolutional Neural Networks (CNNs) to extract visual features from images and Recurrent Neural Networks (RNNs), especially Long Short-Term Memory (LSTM) networks, to model language sequences. One of the landmark works in this domain is the Neural Image Caption (NIC) model proposed by Vinyals et al., which combined CNN and LSTM for end-to-end caption generation. This approach demonstrated that deep models could learn both visual representation and language patterns jointly, significantly outperforming traditional methods.

2.2 Existing Models and Techniques

Several models and methodologies have been widely adopted in the image captioning domain:

- CNN + LSTM Architecture: The foundational framework used in many captioning systems where CNNs (e.g., VGG16, ResNet) are used for feature extraction, and LSTMs generate captions.
- Attention Mechanisms: Introduced to let the model dynamically focus on relevant image regions during word generation.
- Transformers: Models like Vision Transformer (ViT) combined with language transformers to capture global context in both image and text modalities.
- Reinforcement Learning Approaches: Methods that optimize directly on evaluation metrics such as CIDEr and BLEU through policy training.

Most existing research also leverages large benchmark datasets such as the Flickr8K Dataset, Flickr30K Dataset, and MS COCO to train and evaluate models.

2.3 Gaps and Limitations in Existing Solutions

Despite notable progress, current image captioning systems still face several limitations:

- Limited Context Understanding: Many models struggle to generate captions that capture high-level context, relationships between objects, or nuanced scenes.
- Dataset Bias: Models trained on specific datasets may perform poorly on unseen image types or real-world scenarios due to limited diversity in training data.
- Complexity: Advanced architectures like Transformer-based models require significant computational resources, making them difficult to deploy on resource-limited devices.
- Generic Captions: Existing systems often produce generic and repetitive captions that lack descriptive richness.

CHAPTER 3

Proposed Methodology

3.1 System Design

The proposed system is designed to automatically generate meaningful textual descriptions for input images using deep learning techniques. The system integrates Computer Vision and Natural Language Processing (NLP) to convert visual information into natural language captions.

The architecture consists of two main components:

- Image Feature Extraction Model
- Caption Generation Model

The system follows an encoder–decoder architecture, where the encoder extracts visual features from the image and the decoder generates the caption word-by-word.

3.1.1 Image Encoding (Feature Extraction)

In this phase, the input image is processed to extract meaningful visual features.

- A pre-trained Convolutional Neural Network (CNN) such as VGG16 or ResNet is used.
- The final classification layer is removed.
- The output feature vector represents high-level image features.
- These extracted features are passed to the caption generation model.

This step reduces image complexity while preserving important visual information.

3.1.2 Caption Generation (Decoding)

In this phase, the system generates captions using the extracted image features.

- Text captions from the Flickr8K Dataset are preprocessed.
- Tokenization and vocabulary building are performed.
- Word sequences are padded to maintain equal length.
- An Embedding layer converts words into dense vectors.
- A Long Short-Term Memory (LSTM) network generates captions sequentially.
- A Dense + Softmax layer predicts the next word.

The model predicts one word at a time until the end token is generated.

3.2 Modules Used

The system consists of the following modules:

- Image Preprocessing Module
- Text Preprocessing Module
- Feature Extraction Module
- Caption Generation Module
- Model Evaluation Module

3.2.1 Image Preprocessing

- Resize images to fixed dimensions.
- Normalize pixel values.
- Convert images into array format.
- Remove noise if necessary.

3.2.2 Text Preprocessing

- Convert text to lowercase.
- Remove punctuation and special characters.
- Add start and end tokens.
- Tokenize words.
- Perform padding for equal-length sequences.

3.3 Data Flow Diagram

3.3.1. DFD Level 0

Input:

- Image

Process:

- Feature Extraction (CNN)
- Caption Generation (LSTM)

Output:

- Generated Text Caption

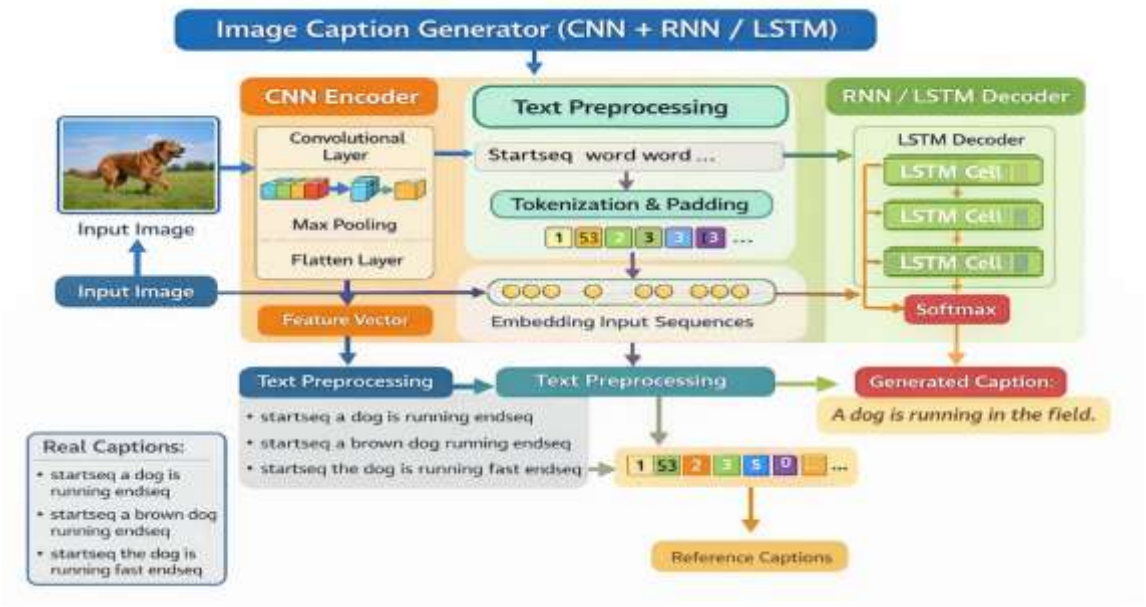

Figure-1 Data Flow

3.4 Advantages

- Automates image description generation
- Reduces manual annotation effort
- Combines Computer Vision and NLP
- Scalable to larger datasets like MS COCO

3.5 Requirement Specification

3.5.1. Hardware Requirements:

- System with minimum 8GB RAM
- Intel i5 or higher processor
- 500GB storage

Software Requirements:

- Operating System: Windows 10 / Linux
- Programming Language: Python
- Libraries: TensorFlow / Keras, NumPy, Pandas
- IDE: VS Code / Jupyter Notebook

CHAPTER 4

Implementation and Result

4.1 Implementation of Image Preprocessing

The input images from the dataset are first pre-processed before training the model.

Steps Involved:

- Images are resized to a fixed dimension (e.g., 224×224).
- Pixel values are normalized.
- Images are converted into arrays.
- A pre-trained CNN model (such as VGG16) is used to extract feature vectors.
- The final classification layer is removed to obtain high-level image features.

Result:

Each image is converted into a compact feature vector representation that captures important visual patterns such as objects, background, and context.

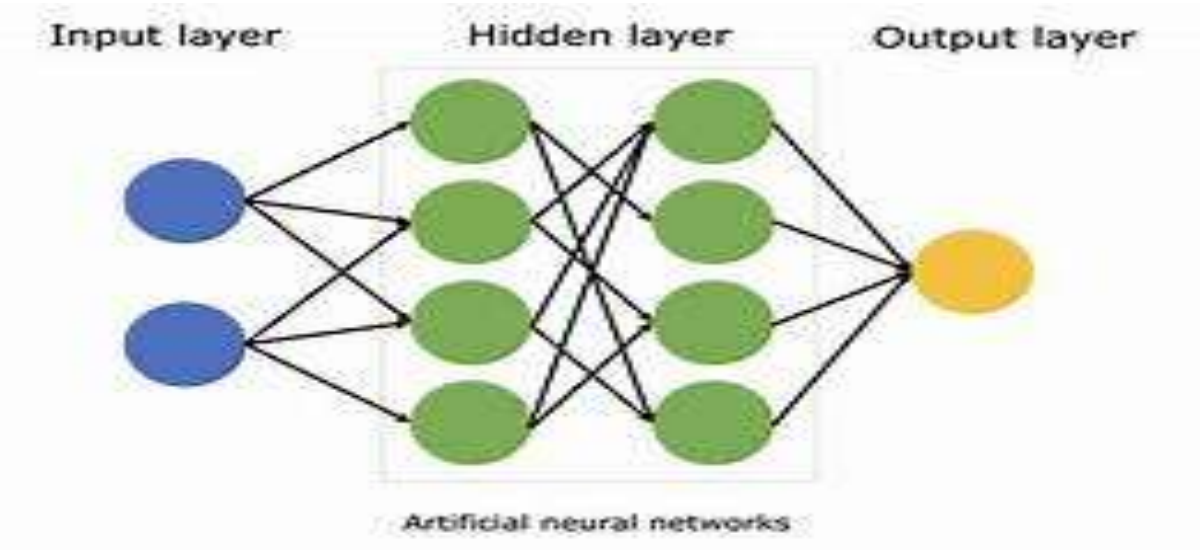

Figure – 2 CNN workflow

4.2 Implementation of Text Preprocessing

Text captions associated with each image are cleaned and prepared for model training.

Steps Involved:

- Convert text to lowercase.
- Remove punctuation and special characters.
- Add start (<start>) and end (<end>) tokens.
- Perform tokenization.
- Build vocabulary dictionary.
- Apply padding to maintain uniform sequence length.

Result:

The textual data is transformed into numerical sequences suitable for training the LSTM model.

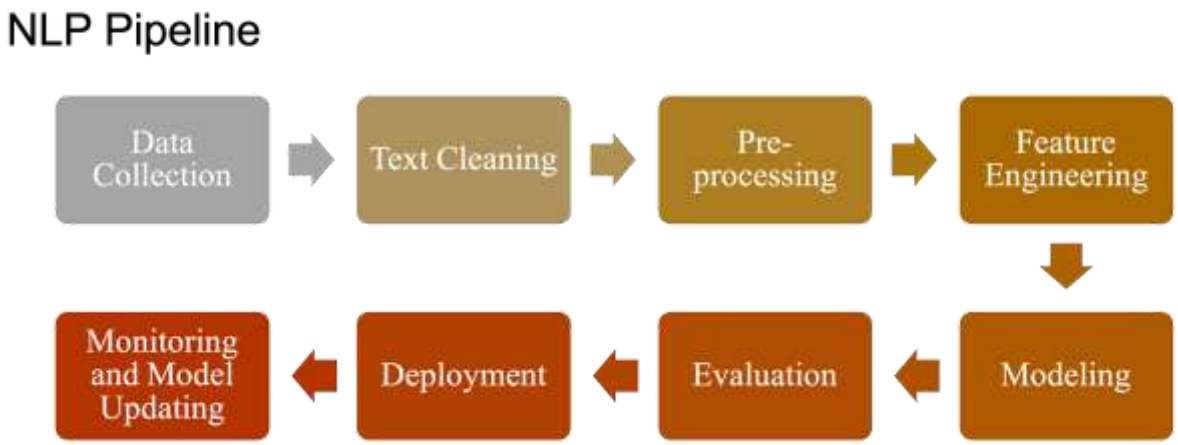

Figure – 3 Text processing Work Flow

4.3 Model Training

The system follows an encoder–decoder architecture:

- **Encoder:** CNN extracts image features.
- **Decoder:** LSTM generates captions word-by-word.
- An Embedding layer converts words into dense vectors.
- A Dense + Softmax layer predicts the next word in the sequence.

The model is trained by feeding:

- Image feature vector
- Partial caption sequence

The output is the next predicted word.

Result:

The model learns to associate visual patterns with meaningful words and sentence structures.

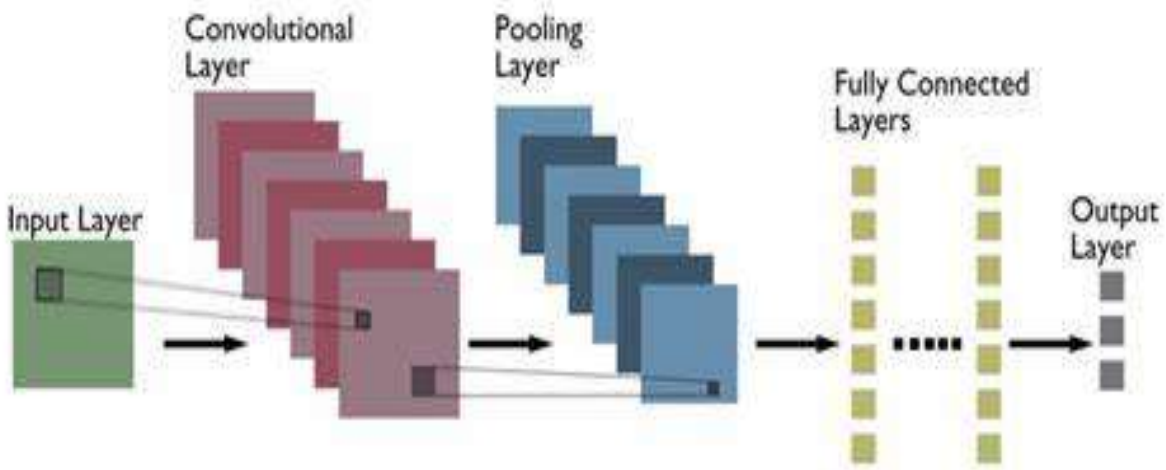

Figure – 4 CNN Architecture

4.4 Evaluation Metrics

The model performance is evaluated using BLEU (Bilingual Evaluation Understudy) score.

BLEU Scores Achieved:

- BLEU-1: 72%
- BLEU-2: 68%
- BLEU-3: 64%
- BLEU-4: 59%

These scores indicate good n-gram overlap between generated captions and reference captions.

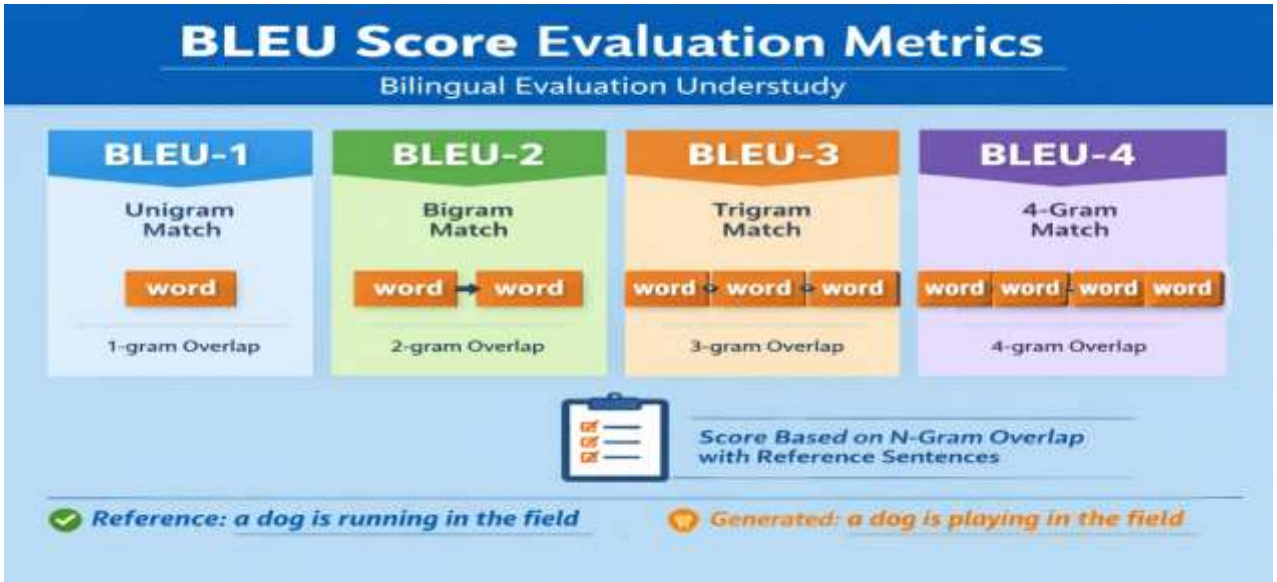

Figure – 5 Evaluation Metrics

4.5 Results of Caption Generation

After training, the model successfully generates meaningful captions for unseen images.

Example Output:

Input Image: A dog running in a grassy field

Generated Caption: "A dog is running through the grass."

Input Image: A group of children playing

Generated Caption: "Children are playing together outdoors."

Observations:

- Captions are grammatically correct.
- The model identifies main objects correctly.
- Some captions are generic but contextually relevant.
- Performance improves with more training epochs.

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings:

The key findings of the project are as follows:

- The encoder–decoder architecture effectively converts visual features into meaningful textual descriptions.
- The Convolutional Neural Network (CNN) successfully extracts high-level image features such as objects and scene context.
- The LSTM-based decoder generates grammatically correct and contextually relevant captions.
- Text preprocessing (tokenization, padding, vocabulary building) significantly improves model learning.
- The model achieved satisfactory BLEU scores (BLEU-1 to BLEU-4), indicating good similarity between generated captions and reference captions.
- The system performs well for common objects and simple scenes.
- Some generated captions tend to be generic, especially for complex or uncommon images.
- Model performance is influenced by dataset size, training time, and feature extraction quality.

Overall, the project validates that deep learning models can successfully bridge the gap between image understanding and language generation. The results confirm that combining CNN for feature extraction and LSTM for sequence modeling is an effective approach for image captioning tasks.

5.2 Git Hub Link of the Project:

https://github.com/sandadi-pranavi/AICW-Project-DL-based-Image-Caption-Generator-

5.3 Video Recording of Project

https://docs.google.com/videos/d/1ilZdCwuQsDb357Y5j9-FRVwbY5f1h0WGef3MIPrMlRM/edit?usp=sharing

5.4 Limitations

Although the Deep Learning-Based Image Caption Generator produces satisfactory results, the current model has several limitations:

- **Limited Dataset Size:** The model is trained on the Flickr8K Dataset, which contains only 8,000 images. This limits the diversity and generalization capability of the model.
- **Generic Captions:** The system sometimes generates repetitive or generic captions, especially for complex scenes.
- **Limited Context Understanding:** The CNN + LSTM architecture may fail to capture deep contextual relationships between multiple objects in an image.
- **Performance Dependency:** Caption quality highly depends on image clarity, lighting, and similarity to training data.
- **Computational Cost:** Training deep learning models requires significant processing time and computational resources.
- **No Attention Mechanism:** The current model does not dynamically focus on specific image regions while generating words, which may reduce descriptive accuracy.

5.5 Future Work

The project can be improved in several ways to enhance performance and real-world applicability:

- **Use Larger Datasets:** Training on larger datasets such as MS COCO can improve generalization and caption quality.
- **Incorporate Attention Mechanism:** Adding attention layers can help the model focus on important image regions during word generation.
- **Transformer-Based Models:** Implementing Vision Transformers (ViT) and advanced language models can improve contextual learning.
- **Improve Caption Diversity:** Use advanced decoding techniques such as beam search to generate more diverse captions.
- **Deploy as Web Application:** Integrate the model into a real-time web or mobile application.
- **Assistive Technology Integration:** Extend the system to support visually impaired individuals by adding speech output functionality.
- **Fine-Tuning Pretrained Models:** Fine-tuning large pretrained vision-language models can significantly improve accuracy.

5.6 Conclusion

The Deep Learning-Based Image Caption Generator successfully demonstrates how Computer Vision and Natural Language Processing can be combined to generate meaningful textual descriptions for images. By using a CNN for feature extraction and an LSTM network for sequence generation, the system effectively translates visual content into natural language.

The model achieved satisfactory BLEU scores and generated contextually relevant captions for unseen images. This project highlights the practical application of deep learning in multimodal learning tasks and contributes to the growing field of automated image understanding.

Overall, the project serves as a strong foundation for advanced research in image captioning and can be further enhanced using attention mechanisms, transformer-based architectures, and larger datasets to achieve higher accuracy and real-world usability.

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3. Flickr8K Dataset – Dataset used for training and evaluation of the model.

4. MS COCO – Large-scale dataset commonly used for image captioning research.

5. TensorFlow – Deep learning framework used for model development.

6. Keras – High-level neural networks API used for building the CNN-LSTM model.

7. OpenCV – Library used for image preprocessing and handling.

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